

Here's a complete walkthrough to create:

1. **A Simple Django CRUD Project**
2. **A Basic AI Application (e.g., Text Sentiment Analyzer)**

Both are runnable on **Windows 11** and **Ubuntu 22.04**.

✔ Project 1: Simple Django CRUD Application

□ Overview

A web app to manage Books with title, author, and description.

□ Directory Structure

```
django_crud_project/
├── manage.py
├── db.sqlite3
├── books/
│   ├── migrations/
│   ├── templates/
│   │   └── books/
│   │       ├── book_form.html
│   │       ├── book_list.html
│   │       ├── book_detail.html
│   │       └── book_confirm_delete.html
│   ├── __init__.py
│   ├── admin.py
│   ├── apps.py
│   ├── models.py
│   ├── urls.py
│   └── views.py
└── django_crud_project/
    ├── __init__.py
    ├── asgi.py
    ├── settings.py
    ├── urls.py
    └── wsgi.py
```

□ Step-by-Step Instructions

1. Create and Activate Virtual Environment

```
# Windows
python -m venv venv
```

```
venv\Scripts\activate
```

```
# Ubuntu  
python3 -m venv venv  
source venv/bin/activate
```

2. Install Django

```
pip install django
```

3. Create Project and App

```
django-admin startproject django_crud_project  
cd django_crud_project  
python manage.py startapp books
```

4. Update settings.py

```
INSTALLED_APPS = [  
    ...  
    'books',  
]
```

□ Code

books/models.py

```
from django.db import models  
  
class Book(models.Model):  
    title = models.CharField(max_length=100)  
    author = models.CharField(max_length=100)  
    description = models.TextField()  
  
    def __str__(self):  
        return self.title
```

books/admin.py

```
from django.contrib import admin  
from .models import Book  
  
admin.site.register(Book)
```

books/urls.py

```
from django.urls import path  
from . import views
```

```
urlpatterns = [
    path('', views.book_list, name='book_list'),
    path('book/<int:pk>/', views.book_detail, name='book_detail'),
    path('book/new/', views.book_create, name='book_create'),
    path('book/<int:pk>/edit/', views.book_update,
name='book_update'),
    path('book/<int:pk>/delete/', views.book_delete,
name='book_delete'),
]
```

django_crud_project/urls.py

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('books.urls')),
]
```

books/views.py

```
from django.shortcuts import render, get_object_or_404, redirect
from .models import Book
from .forms import BookForm

def book_list(request):
    books = Book.objects.all()
    return render(request, 'books/book_list.html', {'books': books})

def book_detail(request, pk):
    book = get_object_or_404(Book, pk=pk)
    return render(request, 'books/book_detail.html', {'book': book})

def book_create(request):
    form = BookForm(request.POST or None)
    if form.is_valid():
        form.save()
        return redirect('book_list')
    return render(request, 'books/book_form.html', {'form': form})

def book_update(request, pk):
    book = get_object_or_404(Book, pk=pk)
    form = BookForm(request.POST or None, instance=book)
    if form.is_valid():
        form.save()
        return redirect('book_list')
    return render(request, 'books/book_form.html', {'form': form})

def book_delete(request, pk):
    book = get_object_or_404(Book, pk=pk)
```

```
if request.method == 'POST':
    book.delete()
    return redirect('book_list')
return render(request, 'books/book_confirm_delete.html', {'book':
book})
```

books/forms.py

```
from django import forms
from .models import Book

class BookForm(forms.ModelForm):
    class Meta:
        model = Book
        fields = ['title', 'author', 'description']
```

HTML Templates in books/templates/books/

- book_list.html, book_form.html, book_detail.html, book_confirm_delete.html (Let me know if you want all template code.)

🔧 Run the App

```
python manage.py makemigrations
python manage.py migrate
python manage.py runserver
```

Open browser: <http://127.0.0.1:8000/>

✔ Project 2: AI App – Text Sentiment Analysis

□ Overview

Input a text → Output: Positive / Neutral / Negative (using TextBlob)

□ Directory Structure

```
ai_sentiment_project/
├── manage.py
├── db.sqlite3
├── sentiment/
│   └── templates/
│       └── sentiment/
```

```
├── index.html
├── __init__.py
├── admin.py
├── apps.py
├── views.py
├── urls.py
└── ai_sentiment_project/
    ├── __init__.py
    ├── settings.py
    ├── urls.py
    └── wsgi.py
```

□ Setup

```
pip install django textblob
python -m textblob.download_corpora
```

□ Code

sentiment/views.py

```
from django.shortcuts import render
from textblob import TextBlob

def index(request):
    sentiment = None
    if request.method == 'POST':
        text = request.POST.get('text')
        blob = TextBlob(text)
        polarity = blob.sentiment.polarity
        if polarity > 0:
            sentiment = "Positive"
        elif polarity < 0:
            sentiment = "Negative"
        else:
            sentiment = "Neutral"
    return render(request, 'sentiment/index.html', {'sentiment':
sentiment})
```

sentiment/urls.py

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.index, name='index'),
]
```

ai_sentiment_project/urls.py

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('sentiment.urls')),
]
```

sentiment/templates/sentiment/index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Sentiment Analyzer</title>
</head>
<body>
    <h2>Enter text to analyze sentiment</h2>
    <form method="post">
        {% csrf_token %}
        <textarea name="text" rows="4" cols="50"></textarea><br><br>
        <input type="submit" value="Analyze">
    </form>
    {% if sentiment %}
        <h3>Sentiment: {{ sentiment }}</h3>
    {% endif %}
</body>
</html>
```

🔧 Run the App

```
python manage.py makemigrations
python manage.py migrate
python manage.py runserver
```

Visit: <http://127.0.0.1:8000/>

✔ Works on Both

Feature	Windows 11	Ubuntu 22.04
Python + Django	✔	✔
TextBlob + AI	✔	✔
SQLite (default DB)	✔	✔

Would you like a downloadable ZIP folder of both projects or want to turn this into a GitHub repository?